Curriculum Vitae - Miguel Mascarenhas Filipe

Lisbon, Portugal
E-mail: miguel.filipe+blog@gmail.com
Web: blog.mfilipe.eu | linkedin.com/in/miguelmfilipe | 13 US patents
European Union citizen

Summary

I'm a Principal Engineer with 23 years of professional experience across start-ups, scale-ups and leading software corporations in the US, UK and Portugal. I dive deep and take an end-to-end perspective. I'm always looking for hard problems to solve and high-quality solutions.

My career centres on databases, query engines and storage systems at scale. At AWS I was on the DynamoDB launch team and was the main author of provisioned throughput. At Dune I've led DuneSQL, the table catalog service and DuneSQL Control Plane (DuneCP), the event-driven control plane that governs the lakehouse.

I have a strong affinity for simple, resilient and performant systems. I'm both a Maker and a Manager. Problems are waiting to be solved.

Professional Experience

Tech Lead, Principal Engineer April 2022 - Present

Dune, the source of truth for onchain data: 115+ chains, 1.5M+ datasets, petabyte-scale lakehouse.

- Led technical direction and delivery across multiple teams on projects requiring more than 12 person-months; established the engineering incident handling process, architecture review forum and architecture decision record (ADR) practice

Notable achievements:

- **DuneSQL** (query engine for blockchain data):
  - Tech lead for DuneSQL's launch, overall design and workstreams; led delivery and evolution with a team of engineers
  - Set the technical direction for the Kubernetes operator, query-execution front end and cluster autoscaling; defined the scope of the Trino fork
  - Designed workload-aware query routing that moved ~50% of production OLAP queries to a single-node deployment; the broader DuneSQL efficiency initiative cut infrastructure costs by 26% and improved gross margin by 22%
- **Catalog service and DuneCP** (control plane):
  - Led the catalog service behind Dune's data explorer and table search, covering materialised views, storage quotas and credit enforcement
  - Designed and built **DuneCP**, an event-driven control plane that unifies maintenance and lifecycle work across roughly one million lakehouse tables per day
  - Designed Datashares V2 for asynchronous, change-driven replication between data lakes
- **Dune API and Query Execution Service**:
  - Built the public Dune API from the first commit through launch (Go), with an OpenAPI-first design, authentication, CSV and Parquet streaming, pagination and metering
  - Built the Query Execution Service with idempotent submission, priority queues and per-tier workers; migrated result storage to object storage in shadow mode with no user-visible impact
  - Built APIs with filtering and pagination over 50 GB Parquet query results using DuckDB

Principal Software Engineer February 2019 - April 2022

Unbabel, an AI-powered human translation platform.

- Chief architect for the Unbabel translation pipeline and AI cluster; highest-ranked engineer in a 110-person R&D division, reporting to the CTO
- Engineering leadership in strategy, project planning, coordination, mentoring and interviewing

Notable achievements:

- Individual Contributor:
  - Designed, led and co-authored the complete re-architecture of the distributed translation pipeline
  - Designed and led *Flowrunner*, our workflow engine for executing and operating every type of translation pipeline
  - Designed and led *maestro*, the monorepo for all NLP processing logic at Unbabel
- Leadership:
  - Led development of the LanguageOS APIs
  - Raised engineering rigour by introducing RFCs, runbooks and shared application metrics libraries
  - Led the adoption of automated end-to-end and feature testing for complex distributed microservice deployments
  - Interim site reliability engineering (SRE) lead: deployed multi-AZ AWS EKS clusters and introduced autoscaling and Terraform

Lead Software Engineer February 2016 - February 2019

Citymapper, an award-winning urban mobility mobile app.

- Lead programmer and chief architect for SmartRide, a bus-taxi hybrid transit service
- Led distributed systems design, development and operations; formerly led SRE

Notable achievements:

- Individual Contributor:
  - Designed, led and co-authored the SmartRide system and its APIs
  - Designed and co-authored a distributed, fault-tolerant HTTP proxy for isolation from external faults
  - Removed numerous single points of failure with a distributed request-submission framework built on DynamoDB, with service discovery, retries and backoff
  - Automated and codified the main infrastructure in CloudFormation, covering dozens of dynamically scaled, sharded microservices
- Leadership:
  - Launched SmartRide from idea to vehicles picking up passengers on London roads in 5 months, with a team of 6 engineers
  - Established SRE practices: better production metrics and visibility, with an emphasis on postmortems
  - Introduced Go as a core programming language; mentored and promoted engineers from junior to senior

Senior Software Engineer November 2014 - January 2016

Microsoft, Application & Systems Group, Aria team (large-scale data ingestion and analytics).

- Developed Aria's distributed data ingestion pipeline, with a focus on scalability and robustness

Notable achievements:

- Delivered data compression at rest and in transit in 4 months, saving more than $12 million per year
- Improved pipeline scalability by more than 2x while reducing end-to-end latency by 90%

Software Engineer - II February 2011 - October 2014

Amazon Web Services, NoSQL Database Services, DynamoDB.

- Contributed to all major components of DynamoDB as a member of the launch team

Notable achievements:

- Main author and implementer of DynamoDB's provisioned throughput
- Co-author of 13 US patents on admission control, workload and performance metering, and throughput maximisation with quality of service (QoS)
- Won the AWS Operational Excellence award for reducing latency and fleet costs through garbage collection tuning (tp99 latency reduced by 33%)

Team Lead, Software Engineer April 2008 - January 2011

SAPO, a Portuguese web portal.

- Led seven engineers working on SAPO's contextual advertising system

Notable achievements:

- Redesigned the database schema with zero downtime and worked extensively with MySQL replication

Software Engineer August 2006 - March 2008

Altitude Software, a provider of call-centre software.

- Developed C# and C++ software for a distributed interactive voice response (IVR) framework

Software Developer May 2005 - July 2006

Master's thesis project, Center for Petroleum Reservoir Modeling, Technical University of Lisbon.

- Developed parallel seismic-modelling algorithms for Linux high-performance computing (HPC) clusters using Message Passing Interface (MPI)

Unix Systems Administrator February 2003 - April 2005

Technical University of Lisbon, Computer and Software Engineering Department.

- Administered Solaris and Linux servers with about 7,000 accounts, plus 150 Linux workstations

Education

September 1999 - July 2006
Master of Engineering (MSc) in Computer and Software Engineering
Instituto Superior Técnico (https://tecnico.ulisboa.pt/en/) (Technical University of Lisbon)

Talks

Slides and recordings at blog.mfilipe.eu/papers

- ACM DEBS 2026 (keynote): DuneCP, an event-driven control plane for a blockchain data platform
- BUIDL Europe 2025: Building with Dune ECHO
- DuckCon #5 (2024): Delighting users with RESTful APIs and DuckDB
- Trino Fest 2023: DuneSQL, a query engine for blockchain data
- 2018: DynamoDB, a view under the hood
- Earlier: Linux Internals (2014), Linux File Systems (2012), How to serve 2500 requests per second (2010)

Technical Experience

- Solid experience designing, developing and operating large-scale distributed systems (10,000+ nodes)
- Professional experience in Go, Python, C/C++, Java and SQL; previously C#
- Databases, engines and table formats: DynamoDB, PostgreSQL, FoundationDB, CockroachDB, Trino, Iceberg, DuckDB and MySQL
- Storage: object-storage-backed lakehouses; tiered NVMe caches and eviction policies; durability and end-to-end integrity
- Good knowledge of Linux kernel internals (VFS, BIO, scheduler, memory manager, filesystems)
- Good knowledge of virtualisation and container technologies (Kubernetes, cgroups, KVM, Docker)
- Good knowledge of hardware architectures and topologies, IP networking and network protocols
- Observability and capacity engineering with Prometheus, Thanos and Grafana; infrastructure as code with Terraform
- Knowledge of x86 assembly

Technical Interests

- High-performance, large-scale distributed applications
- Distributed data stores and filesystems; consensus, replication and their failure modes
- Highly reliable and highly durable storage systems; deterministic simulation testing
- Full end-to-end systems operations: from applications and the software stack through the OS and hardware topology to data-centre design
- Operating-system internals, computer hardware design and architecture